PillPlan

Product Requirements Document

A medicine schedule maker that turns a list of medicines into a clear picture chart a patient can follow.

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Course	Pharmacy class project
Live	`https://pillplan-cela.netlify.app`

> **In one sentence.** At the pharmacy counter, the apoteker types the prescription signa (for example `3 dd 1 tab p.c.`), PillPlan checks the plan and turns it into one clear picture chart, a counseling view and phone reminders, in English or Indonesian.

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1 Problem

Patients who take several medicines often take them at the wrong time, with the wrong food, or twice by accident. The risk is highest for older adults and for people with low health literacy. Labels such as *"take 1 tablet twice daily"* leave the patient to work out the times themselves.

Research gives a clear direction:

- The **Universal Medication Schedule (UMS)** showed that a fixed grid of times improves understanding for every literacy level. PillPlan keeps the grid but not the four day parts: current pharmacy practice spreads doses evenly, so "3 dd 1" means **every 8 hours** (for example 07.00, 15.00 and 23.00), not morning, noon and evening. Even gaps keep blood levels steady, which matters most for antibiotics.
- Patients recognise a medicine by its **sediaan** first: tablet, capsule, syrup or suspension. The exact pill colour matters less than knowing that a suspension must be shaken and measured.
- The **AHRQ Pill Card** shows that a simple table with a row per medicine, sun and moon icons and a "what it is for" column is easier to follow than text.
- Good instructions use **numbers instead of words**, put **each dose on its own line**, and use **plain words** for the purpose.

In Indonesia the apoteker reads the signa, writes the etiket and gives drug information and counseling at handover, as required by Permenkes 73/2016. An Indonesian study found that elderly patients understood the etiket in 99% of cases, yet only 54% of those with 5 medicines took all of them. Understanding one label is not the problem; organizing the whole day is.

The AHRQ card is a Word template. It is slow to fill in, it does not check anything, and it cannot help during counseling. PillPlan keeps the proven layout and adds speed, checks and interaction.

2 Why not a reminder app?

A classmate is building a pill reminder app. The two products solve different problems and work well together.

	Reminder app	AHRQ pill card	PillPlan
Who uses it	Patient at home	Pharmacist, by hand	Apoteker at the counter
When	Every day	At handover	At handover and counseling
Data entry	Patient types it	Typed in Word	Signa or catalog, prefilled
Checks the plan	No	No	Daily maximum, duplicates, timing
Works without a phone	No	Yes, paper	Yes, paper and screen
Reminders	Yes	No	Yes, through the phone calendar
Language	Varies	English	English and Indonesian

Reviews of reminder apps for older adults find mixed results: many users stop using the app, and studies with cognitively impaired users showed no benefit. PillPlan therefore puts the effort where the plan is made, and hands reminders to the phone calendar the patient already has.

3 Users

User	Need	What PillPlan gives
Apoteker	Turn a prescription into a clear plan in under 2 minutes	Signa entry, checks, one-page print, counseling mode
Pharmacy student	Practice reading signa and counseling	Live signa preview, safety notes, counseling mode
Caregiver	Manage a parent's medicines	Plain words, tracker, saved in the browser
Patient	Know what to take and when	Picture chart, pill drawings, warning stickers

3.1 Personas

Sari, 21, pharmacy student. Practices counseling for an exam. Wants to see if her dosing is sensible and to explain each medicine one at a time.

Mr Budi, 72, patient. Takes 4 medicines for diabetes, blood pressure and cholesterol. Reads large print best. Keeps the chart on the fridge.

4 Goals and non-goals

Goals

1. A correct chart for 4 to 6 medicines in **under 2 minutes**.
2. The chart fits on **one A4 page** and reads well in black and white.
3. The app is **fast, light and private**: no account, no server, no upload.
4. Simple checks catch common mistakes such as hidden paracetamol.

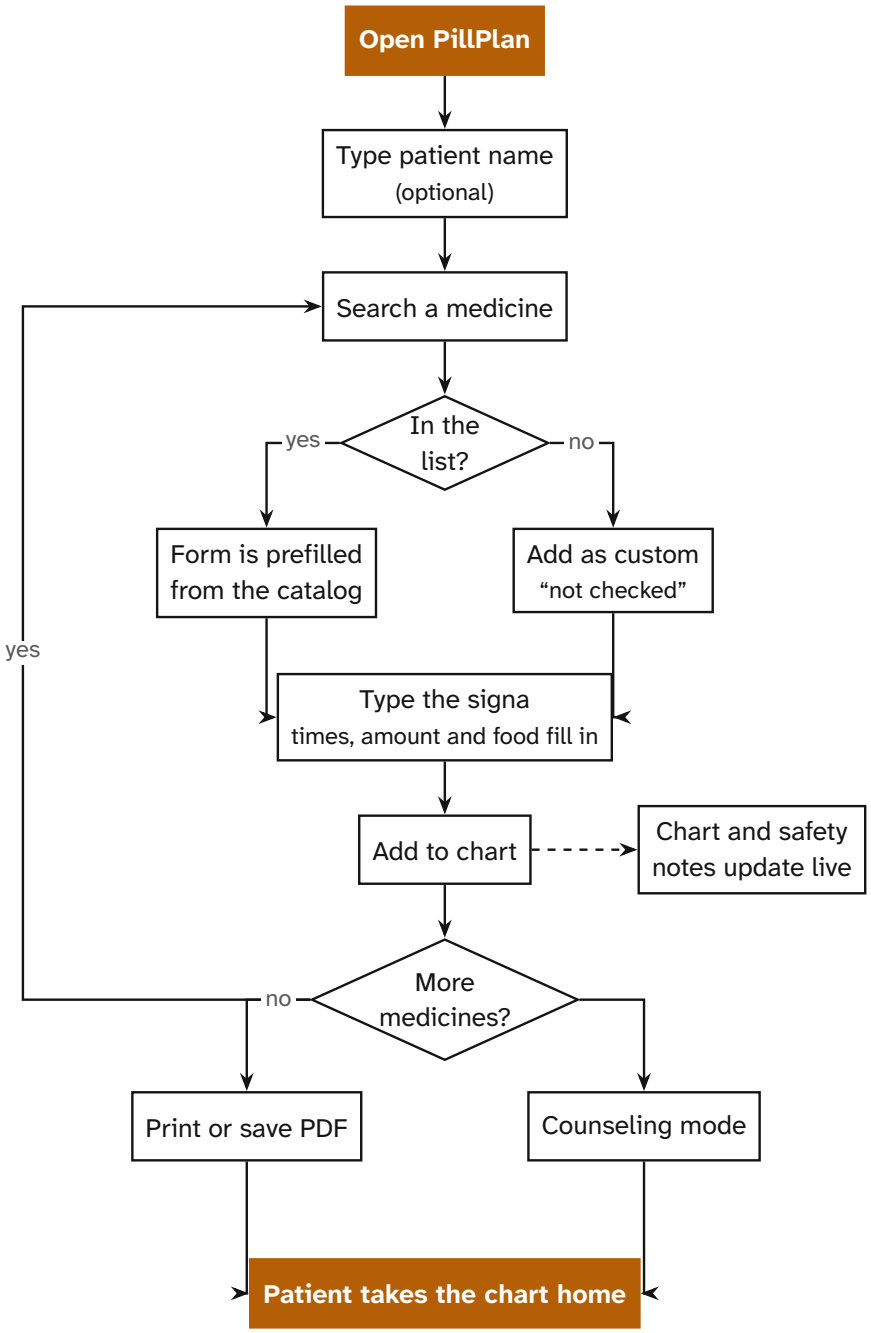
Non-goals

- Accounts, cloud sync or sharing between devices.
- A full drug interaction checker.
- Prescribing advice. PillPlan never says a plan is “safe”.

5 Features and acceptance criteria

Feature	Accepted when
Drug search	Generic, brand or Indonesian name (“parasetamol”, “CTM”) lists matches at once. Picking one fills strength, usual time, food rule, warnings and advice.
Signa entry	Typing `3 dd 1 tab p.c.` or `2x1 sesudah makan` fills the interval (every 8 or 12 hours), the first dose, amount and food, shows a plain preview, and lists any word it did not understand.
Own drugs	Any drug can be typed. It is tagged “not checked” and remembered on the device as “My drugs” for next time.
Language	One switch changes the whole app, the chart and the print between English and Indonesian, including number format (`0,5 mg`).
Phone reminders	One button downloads a calendar file with a daily reminder for each dose time.
Dose clock	A 24-hour dial shows the doses evenly around the day. Dragging the first dose, or using the arrow keys, moves every dose and keeps the interval.
Sediaan	Tablet, capsule, syrup or suspension, drawn as a simple picture. Liquids show “measure with the dosing spoon”, suspensions also “shake well”. Counseling mode shows a 3D model the patient can turn.
Welcome film	On the first visit a 20 second film tells the problem and the solution, then the app opens. It is rendered ahead of time, so the site ships no video library.
Medicine form	Amount in half steps, every 24, 12, 8 or 6 hours from a first dose or “when needed” with a daily limit, food rule, warning stickers, purpose, note and sediaan. Errors are shown next to the field.
Live chart	Each medicine is one row, each dose time (such as 07.00) is one column, doses are numbers. When-needed medicines sit in their own box.
Safety notes	Daily total per ingredient across products, duplicate ingredient, duplicate class and time-of-day hints.
Reorder	Drag with mouse or touch, or move with the keyboard. Screen readers hear the new position.
Spotlight	Tapping a time highlights only the doses at that time.
Tracker	A 7-day blister pack. Tapping a bubble marks a dose as taken and is saved.
Counseling mode	Full screen, one medicine at a time, big text, arrow keys.
Saving	Autosave in this browser, a switch to turn it off, clear all data, export and import a file.
Print	Only the chart prints, on A4, with names and doses at 14 pt or larger.

6 User flow



7 The chart

The chart keeps the UMS grid, with one column per clock time instead of per part of the day. The drawing below shows the layout with the example patient used in demos.

Medicine	07.00	15.00	19.00	21.00
Metformin 500 mg for diabetes	1		1	
Amlodipine 5 mg for blood pressure	1			
Simvastatin 20 mg for cholesterol				1

When needed

Paracetamol 500 mg. Take 2 tablets. No more than 4 times a day. **No alcohol**

8 Requirements

8.1 Performance

- Initial JavaScript at most 110 KB gzip. Measured: **93.6 KB**.
- Fonts at most 60 KB. Measured: **43.9 KB**.
- The drug list and counseling mode load as separate chunks (**21.5 KB** together).
- Lighthouse 95 or higher in every category. Measured on mobile: **100, 100, 100 and 100** for performance, accessibility, best practices and SEO.

8.2 Accessibility

- WCAG 2.2 AA: contrast, visible focus, touch targets of 44 px or more, full keyboard use.
- Atkinson Hyperlegible Next, designed by the Braille Institute for low-vision readers.
- Motion is turned off when the device asks for reduced motion.

8.3 Security and privacy

- A fully static site: no server, no database, no accounts, no third-party scripts.
- Patient data stays in the browser. The export file name never contains the patient name.
- A strict Content Security Policy and other security headers on every response.
- Saved and imported data is validated before use.

8.4 Medical safety

- Every catalog row names its source. Starter data comes from US FDA labels and must be checked against a local formulary.

- Notes say “check with your pharmacist”. Custom medicines are tagged “not checked”.
- Numbers follow ISMP rules: `0.5 mg`, never `.5 mg` or `1.0 mg`.

9 Success metrics

Metric	Target	Status
Time to build the example chart	under 2 minutes	to measure with classmates
Chart fits on A4	1 page	met (print test)
Automated tests passing	all	132 of 132
Initial JavaScript (gzip)	$\leq$ 110 KB	93.6 KB
Catalog medicines	35 or more	43
Lighthouse, every category	$\geq$ 95	100, 100, 100, 100
Classmates who understand the chart	4 of 5	to measure

10 Risks

Risk	Mitigation
Wrong catalog data	Python validation, a source per row, review against the formulary before real use.
User trusts “no notes” as safe	Wording never says safe, and lists what is not checked.
Data loss when the browser is cleared	Export file, and a clear note about where data lives.
Shared computers	“Remember on this device” switch and Clear all data.
Older browsers	View Transitions fall back to normal updates.

11 Evaluation study

A small study for the class, so the value is measured, not only claimed.

11.1 Study 1: does the chart help patients understand?

- **People:** 10 adults who are not pharmacy students (family, friends), 5 aged 50 or older if possible.
- **Material:** the same 5-drug plan shown two ways: (A) five etiket texts such as “3 x sehari 1 tablet sesudah makan”, (B) the PillPlan chart. Half see A first, half see B first.

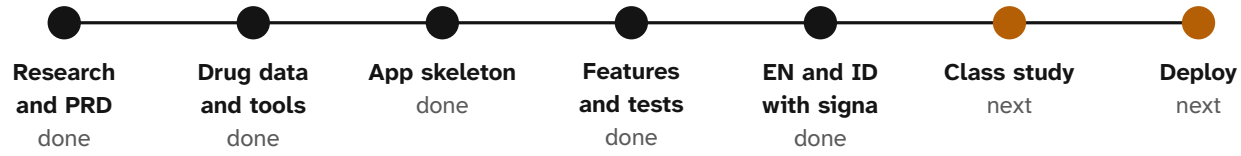
- **Questions:** 6, for example “Which medicines do you take at night?”, “How many tablets at noon?”, “Before or after food for metformin?”
- **Measures:** correct answers out of 6, and seconds to answer all.

11.2 Study 2: is it quick and easy for the apoteker?

- **People:** 5 pharmacy students.
- **Task:** turn 3 sample prescriptions into charts, once by hand (etiket) and once with PillPlan.
- **Measures:** minutes per prescription, errors found by a second student, and the System Usability Scale (SUS, 10 questions, 0 to 100; above 68 is above average).

Result (fill in)	Etiket text	PillPlan
Correct answers (mean of 6)		
Time to answer (seconds)		
Minutes per prescription		
Errors per prescription		
SUS score	not used	

12 Milestones



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